Installation of MetaMask and Study of Spending Ether per Transaction

Network: Ethereum Sepolia (test network) | Wallet: MetaMask | Account 1: 0x28459...01Be4

Step 1: MetaMask extension installed from the Chrome Web Store

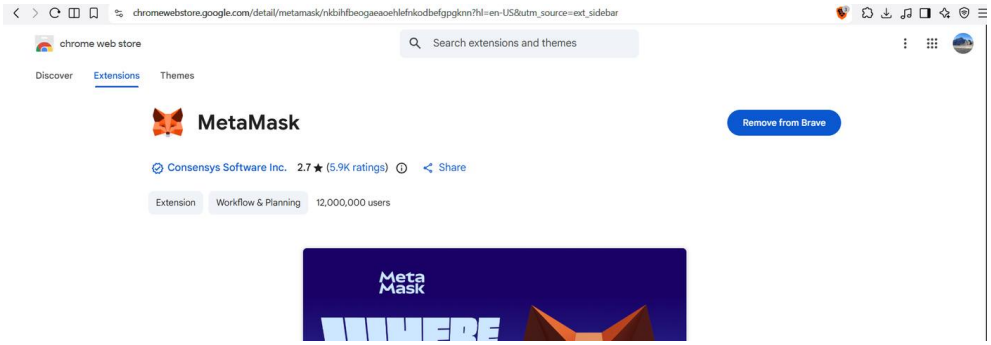
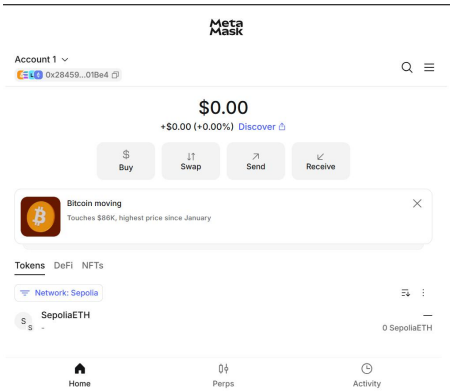


Fig 1: MetaMask (Consensys Software Inc.) installed in the browser

Step 2: Wallet created on Sepolia, test ETH requested from the faucet



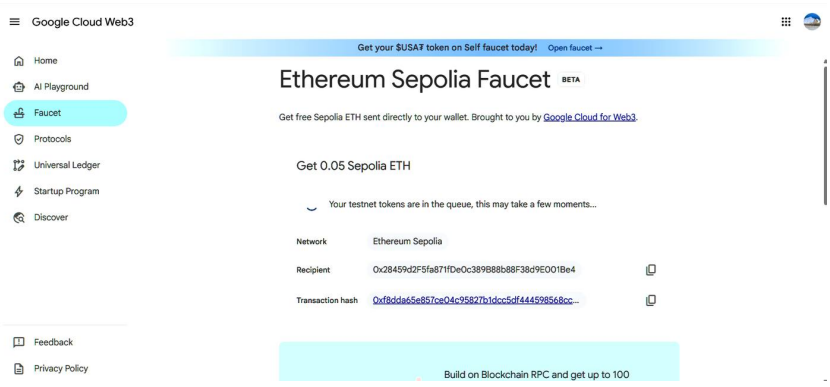


Fig 2: MetaMask wallet (Account 1) on Sepolia, balance 0 SepoliaETH

Fig 3: Google Cloud Web3 faucet sending 0.05 Sepolia ETH to the same address

Step 3: Faucet transaction confirmed, wallet funded

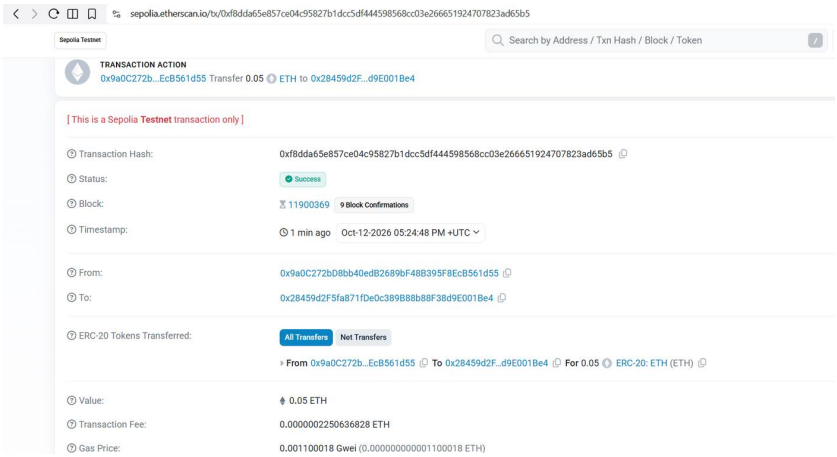


Fig 4: Etherscan: Status Success, value 0.05 ETH, fee 0.0000002250636828 ETH (paid by the faucet)

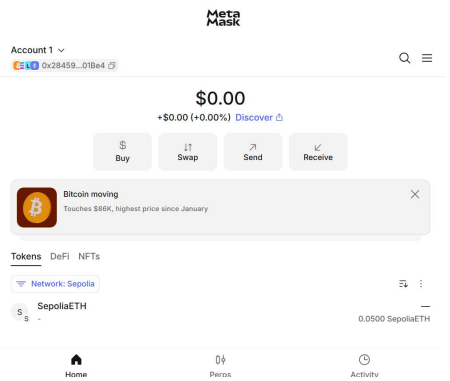


Fig 5: Wallet now shows 0.0500 SepoliaETH

Step 4: Spending Ether: sending 0.01 SepoliaETH

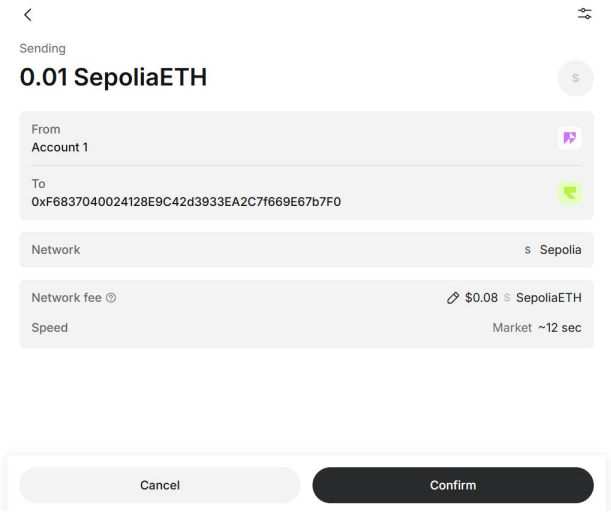


Fig 6: Send screen: amount, From, To, network fee and speed before Confirm

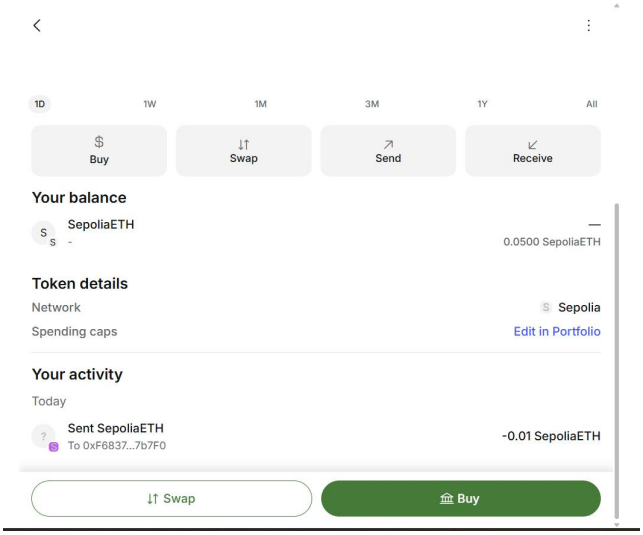


Fig 7: Wallet activity: Sent SepoliaETH, -0.01

Step 5: Fee paid for the 0.01 ETH transfer (Etherscan)

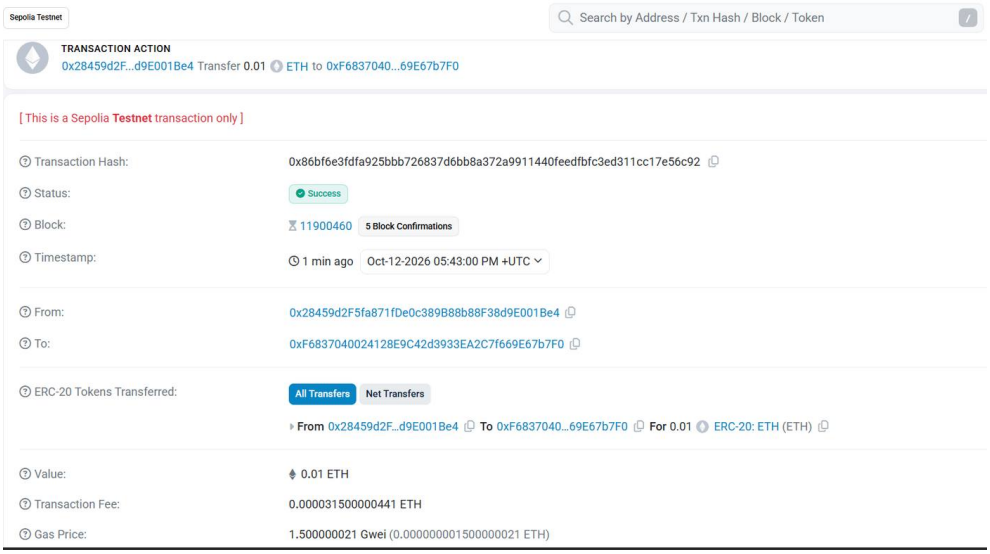


Fig 8: Status Success, value 0.01 ETH, Transaction Fee 0.000031500000441 ETH, Gas Price 1.500000021 Gwei

Step 6: Fee estimates shown by MetaMask for contract transactions

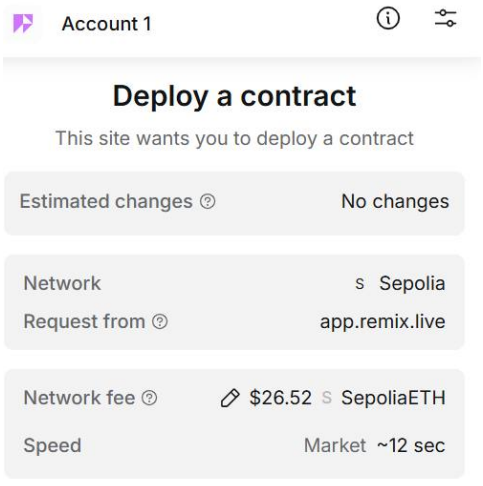


Fig 9: Deploy a contract: network fee estimate $26.52 SepoliaETH (display value)

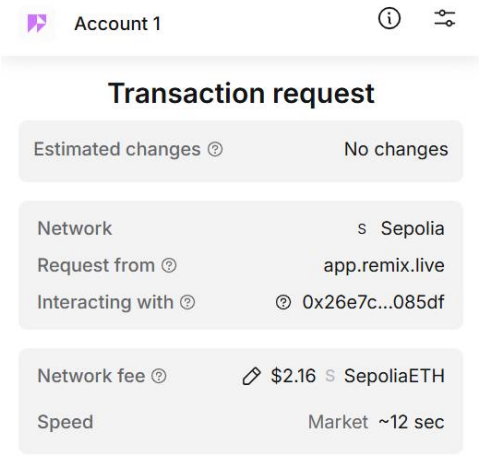


Fig 10: Contract call (addStudent): network fee estimate $2.16 SepoliaETH (display value)

Step 7: Actual fee for the contract call (addStudent)

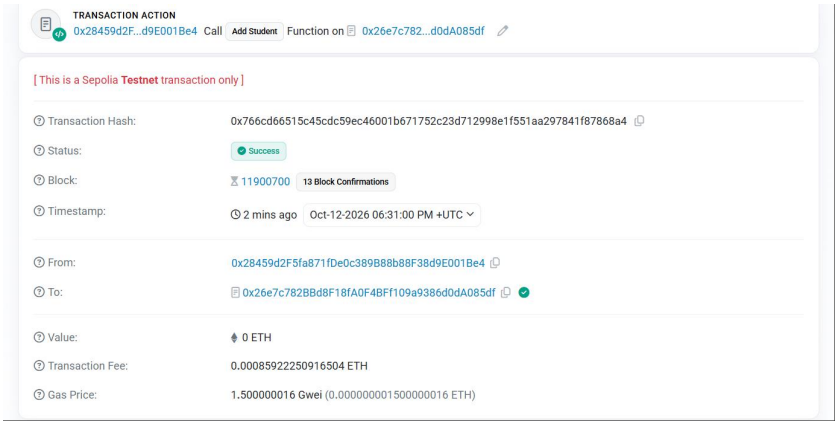


Fig 11: Etherscan: Success, Transaction Fee 0.00085922250916504 ETH, Gas Price 1.500000016 Gwei



Fig 12: MetaMask: Contract interaction, Status Confirmed

Fee comparison

Transaction	Fee (ETH)	Gas price	Gas used (fee / price)
Faucet to wallet (paid by faucet)	0.0000002250636828	0.001100018 Gwei	about 204,600
Send 0.01 ETH (my wallet)	0.000031500000441	1.500000021 Gwei	21,000
addStudent call (my wallet)	0.00085922250916504	1.500000016 Gwei	572,815

Conclusion: Every transaction costs a network fee = gas used x gas price, on top of the amount sent. A plain ETH transfer needs the minimum 21,000 gas, while contract calls such as addStudent use far more gas because they write data to storage. Sepolia ETH is a test token with no real value; dollar amounts in MetaMask are display conversions only.